Intelligent Subtitles: Automatic Adaptation of Subtitle Display Based on Audio and Text Comprehensibility

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Abstract

This work addresses the limitations of conventional subtitle systems through the development of an adaptive approach that displays subtitles on demand based on audio and text comprehensibility. Traditional permanent subtitles frequently lead to visual distraction, image occlusion, and unnecessary reading, while their complete absence can significantly impair understanding of foreign language content. The presented methodology combines nine carefully developed features (word complexity, sentence complexity, word importance, word occurrence, word familiarity, audio complexity, speech rate, ambient volume, and relative volume) with user preferences in a pipeline that leads to selective subtitle display through an Extreme Gradient Boosting model. The quantitative evaluation shows promising results with an F2-score of 0.6122, a balanced accuracy of 0.8064, and a ROC-AUC value of 0.8927. The model demonstrates particular strength in suppressing unnecessary subtitles (precision of 0.97 for the "non-display" class), but shows optimization potential for the precision of the "display" class (0.37). The feature relevance analysis identifies user language level, word familiarity, and word frequency as particularly influential for model decisions. The results confirm the fundamental feasibility of adaptive subtitle systems while simultaneously showing specific directions for future optimizations, particularly regarding training set expansion and feature extraction refinement.

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1 Technical Summary

This work presents a system for dynamic adaptation of subtitle display based on the analysis of audio and text comprehensibility. The goal is to display subtitles selectively and user-centrically to minimize visual distractions and maximize comprehensibility. Due to the lack of comprehensive training data, a pipeline was developed that combines nine central features:

- **Word complexity:** The linguistic difficulty of words is assessed through assignment to CEFR levels (A1 to C2).
- **Sentence complexity:** The structural and semantic complexity of a sentence is estimated using a Pretrained Encoder-Only Model, also known as Pretrained Transformer (PT).
- **Word importance:** Essential words in the context of a sentence are identified by masking them and evaluating their predictability with a PT.
- **Word occurrence:** The frequency of a word in the video is used to avoid redundant subtitles.
- **Word familiarity:** The everyday frequency of words is evaluated.
- **Audio complexity:** Based on the uncertainty assessment of transcriptions extracted by an automatic speech recognition system (ASR), the difficulty of the audio signal is analyzed.
- **Speech rate:** The syllable rate per second is calculated for each sentence.
- **Ambient volume:** The ratio between speech volume and background noise is analyzed to identify poorly understandable words.
- **Relative volume:** The relative volume of a sentence is evaluated.

In addition to the mentioned features, two user self-assessments of English level and audio comprehension are used as input for an Extreme Gradient Boosting Model that was trained to adaptively display or hide subtitles. The model was trained and tested with a dataset of approximately 40,000 labeled words, with a test size of 20%.

The evaluation on the test set yielded a balanced accuracy of 0.8064 and a ROC-AUC value of 0.8927, confirming the model's fundamental ability to distinguish between both classes. Class 1 (display) achieves a recall of 0.73, so a large portion of necessary subtitles is recognized. However, the precision for this class is low at 0.37, indicating a relevant number of falsely displayed subtitles (false positives). For class 0 (non-display), the model achieves a high precision of 0.97, effectively suppressing unnecessary subtitles.

In summary, the work demonstrates the potential of adaptive subtitle systems to reduce visual overload while simultaneously supporting comprehensibility. Future work should focus on improving precision (reducing false positives), optimizing pipeline speed, and expanding the training set with more diverse user profiles and content to increase generalizability and individual adaptability.

2 Motivation and Research Question

Audiovisual media with subtitles play a central role in modern society, both in the field of accessibility for people with hearing impairments (Bundesfachstelle Barrierefreiheit, 2022) and in foreign language acquisition and consumption (Schmidt-Weigand et al., 2017). Current studies show that more than half of viewers regularly use subtitles (Preply, 2022). However, the growing relevance of subtitle systems contrasts with existing limitations: permanent subtitles frequently lead to visual distraction, obscure relevant image elements, and can consume cognitive resources through unnecessary reading (Schmidt-Weigand et al., 2017). Without subtitles, however, understanding of foreign language content is impaired, especially with complex language, fast speaking pace, or disturbing background noise.

Research on adaptive media systems has made significant progress in recent years. Hu et al. (2014) developed dynamic subtitle systems that position subtitles based on the speaker's position. Akahori et al. (2017) and Kurzhals et al. (2020) extended this approach by considering the viewer's gaze position. Gorman et al. (2021) introduced a user-centric system that enables personalized decisions by allowing viewers to determine for which film characters subtitles should be displayed. However, these studies primarily focus on the spatial positioning of subtitles, while adaptive control of subtitle visibility based on actual necessity has been insufficiently researched.

The present work addresses this research gap through the development and evaluation of a system for demand-oriented subtitle fade-in and fade-out. Unlike existing approaches, this system considers not only visual presentation but integrates a comprehensive analysis of audio signal comprehensibility and linguistic complexity in combination with individual user preferences. The central research question is: To what extent can a model based on audio and text features precisely control the adaptive display of subtitles and thereby overcome the disadvantages of conventional subtitle systems?

The work is based on the hypothesis that a combination of audio complexity, text features, and user-specific factors may be sufficient to develop an effective adaptive subtitle system. To test this hypothesis, a model was developed and evaluated using quantitative metrics and user-centric evaluations.

3 Background and Theoretical Foundations

A central problem was the lack of a suitable training set. Such a set would need to contain not only assessments of the difficulty of individual words but also information about the audio and text comprehensibility levels of users. Deep neural networks like Transformers (Vaswani et al., 2017), RNNs (Karpathy, 2015), or LSTMs (Hochreiter and Schmidhuber, 1997) typically require large datasets for training or post-training. Since such a dataset was not available, a pipeline was developed that extracts nine central features to model the comprehensibility of audio and text without an extensive training set: word complexity, sentence complexity, word importance, word occurrence, word familiarity, audio complexity, speech rate, ambient volume, relative volume.

4 Approach, Materials, and Methods

4.1 Pipeline Architecture

The pipeline requires as input the audio of a video as well as the original subtitles. First, manual cleaning occurs, where sounds or irrelevant words are removed and non-English words are marked. In the further processing steps, the nine central features are then extracted. The subtitles are additionally translated into German to later also display bilingual subtitles or exclusively the translation of difficult words. A self-trained Extreme Gradient Boosting Model processes the extracted features and the users' audio and text comprehensibility specifications and makes the final decision about subtitle display. As a result, the pipeline delivers optimized subtitles that are only displayed when they are truly necessary for understanding.

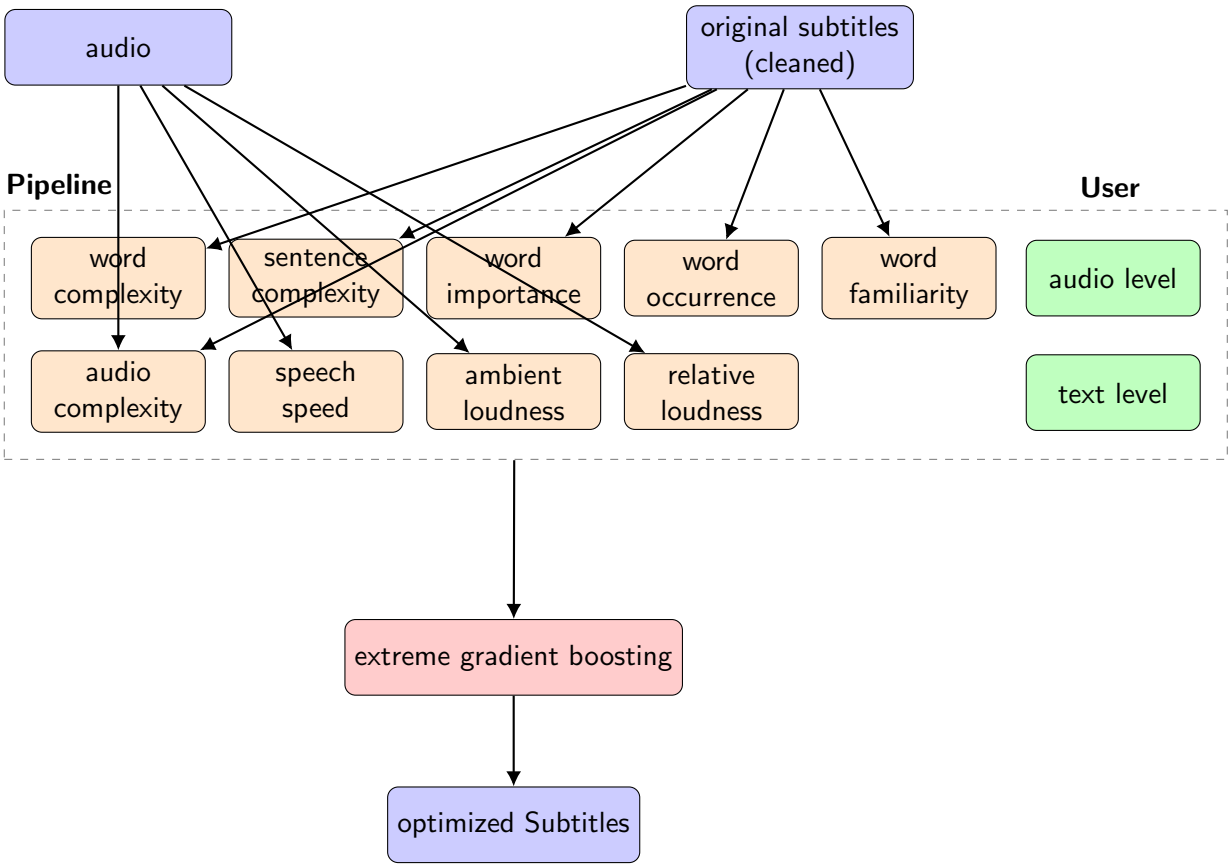


Figure 1: Pipeline structure

4.2 Feature Extraction

4.2.1 Word Complexity

The complexity of a word is calculated based on its CEFR level. A mapping table (see Table 1) defines values between 0 and 1. More complex words receive higher values:

CEFR-Level	word complexity
A1	0.0
A2	0.2
B1	0.4
B2	0.6
C1	0.8
C2	1.0

Table 1: Mapping of CEFR levels to complexity values.

4.2.2 Sentence Complexity

Sentence complexity is calculated using ModernBERT (Warner et al., 2024), a faster and more modern variant of the original BERT (Devlin et al., 2018), to evaluate the difficulty of a sentence. The prediction uncertainty for each word is used:

$$Complexity(s) = \frac{U(s)}{U_{max}} \tag{1}$$

where $U(s)$ is the cumulative prediction uncertainty of sentence s and U_{max} is the maximum value of a sentence defined as maximally complex as a reference to obtain general normalization. The prediction uncertainty $U(s)$ is calculated as follows:

$$U(s) = -\sum_{i=1}^{N} p(w_i) \log p(w_i) \tag{2}$$

where N is the number of words in the sentence.
A sentence thus achieves high complexity values when it contains many rare or unexpected words (e.g., unusual words or complex structures) and low values when the words are predictable (e.g., frequent words or simple structures).

4.2.3 Word Importance

The importance of a word within a sentence is calculated through masking predictions with ModernBERT (Warner et al., 2024). For each sentence, a word is masked, and the probability for the original word is determined:

$$Importance(w_i) = 1 - p(w_i|s). \tag{3}$$

The more unexpected a word is in context, the higher its importance.

4.2.4 Word Occurrence

The frequency of a word in the entire video is normalized as follows:

$$Occurrence(w_i) = \frac{A(w_i)}{A(max)}. \tag{4}$$

where $A(w_i)$ is the frequency of the word in the entire text and $A(max)$ is set to 15 in the current pipeline, as words that appear more frequently in a video are assumed to be sufficiently familiar.

4.2.5 Word Familiarity

Word familiarity is based on the normalized Zipf scale (van Heuven et al., 2014), which quantifies the perceived frequency of words in a language.
The Zipf familiarity is determined and normalized:

$$Familiarity(w_i) = \frac{Zipf\text{-}Familiarity(w_i)}{8} \tag{5}$$

Normalization is performed by division by 8 due to the value range of the Zipf scale, which lies between 0 (very rare) and 8 (extremely frequent).

4.2.6 Audio Complexity

In the first step, the audio is transcribed with Whisper (Radford et al., 2022). A smaller model (whisper-tiny, quantized to 8-bit integer) is used to capture larger uncertainties and outliers. For each word w_i, the uncertainty assessment is calculated as follows:

$$A(w_i) = 1 - p(w_i), \tag{6}$$

where $p(w_i)$ is the probability predicted by the model for the word. Words with high uncertainty thus receive higher values.

4.2.7 Speech Rate

Speech rate is determined by analyzing the syllable rate per second for each sentence. Unlike word-based analysis, entire sentences are considered here, as the speed of individual words can vary too much and does not represent a meaningful metric. For implementation, a larger, better Whisper model is used that provides more precise timestamps for each word.
For each sentence, the syllable rate is calculated as follows:

$$SPS(s) = \frac{\sum_{i=1}^{N} Syllables(w_i)}{Duration(s)} \tag{7}$$

where $SPS(s)$ is the syllables per second for sentence s, $\sum_{i=1}^{N} Syllables(w_i)$ is the sum of syllables of all words in the sentence, and $Duration(s)$ is the temporal length of the sentence in seconds.
For normalization of speech rate, a sigmoid function is used:

$$Speed(s) = \frac{1}{1 + e^{-(SPS(s) - CENTER)/SCALE}} \tag{8}$$

The parameters $CENTER = 5.2$ and $SCALE = 1.2$ are based on well-founded scientific studies. Nitta et al. (2010) were able to demonstrate that the average conversation rate in films and series is about 5.0 syllables per second, while Robb et al. (2004) documented that American adults speak on average at 5.3 syllables per second. Based on these findings, the mean value of 5.2 was set as the $CENTER$ parameter.

The standard deviation of 1.2 was derived from the percentile data of the study, which shows that about 60% of natural speech lies between 4.0 and 6.0 syllables per second.[1]

For the next two features, the audio track is separated into voices and background noise using demucs (Rouard et al., 2023). Demucs is a hybrid approach that combines Fourier and wavelet transformations with deep neural networks to achieve precise source separation.

4.2.8 Ambient Volume

Ambient volume quantifies the ratio between voice and background volume as an objective metric for each word:

$$Ratio(w_i) = \frac{Voice\text{-}Volume(w_i)}{Background\text{-}Volume(w_i)} \tag{9}$$

Volume is calculated as the Root Mean Square (RMS) of the amplitude of the respective audio segment:

$$Volume(segment) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} segment_i^2} \tag{10}$$

A ratio less than 1 indicates that the background is more dominant than the voice, significantly reducing comprehensibility. To particularly emphasize these critical cases for adaptive subtitle display, logarithmic normalization is implemented:

$$Ambient\text{-}Volume(w_i) = 1 - \frac{\log(1 + Ratio(w_i))}{\log(1 + C)} \tag{11}$$

where $C = 4$ serves as a scaling factor.

4.2.9 Relative Volume

Relative voice volume captures the variation in speaker intensity within the entire video.
For each sentence, the average voice volume (RMS) is calculated and normalized relative to the global average of all sentences:

$$Ratio(s) = \frac{RMS(s)}{RMS_{mean}} \tag{12}$$

This ratio is linearly transformed to a value range of 0 to 1, where 0 is defined as minimum and 2 (double average volume) as maximum:

$$Relative\text{-}Volume(s) = \frac{Ratio(s)}{2} \tag{13}$$

The calculation is performed at the sentence level, as natural prosodic variations of individual words would distort measurement at the word level.

[1]The estimation of the standard deviation was made based on the percentile distribution from Figure 3 in Nitta et al. (2010). With approximate normal distribution, the 50th percentile corresponds to about 5.0 SPS, while about 18.7% 4 SPS and 77.2% 6 SPS. This means that approximately ±1 SPS around the mean covers a range of about 19% to 77%, which approximately corresponds to one standard deviation in a normal distribution (with perfect normal distribution, ±1 standard deviation would cover 68%; here we have about 58.5% between 4–6 SPS). This suggests that the standard deviation of speech rate is about 1.0 SPS. For conservative estimation, the $SCALE$ parameter was set to 1.2 SPS.

4.3 Context-Dependent Translation

Each sentence is also translated into German using OPUS-MT (Tiedemann and Thottingal, 2020) to later also display bilingual subtitles. With SimAlign (Jalili Sabet et al., 2020), the original words are aligned with the translated words to consider context-dependent meanings.
Example: Original sentence (English) „The bark of the tree was rough.“German translation: „Die Rinde des Baumes war rau.“
The alignment would assign the word pairs as follows:

- The → Die
- bark → Rinde
- of → des
- the → des
- tree → Baumes
- was → war
- rough → rau

Comment: The word *bark* is a homonym that can mean both *Rinde* and *Bellen*. The translation considers the context of the sentence here.

4.4 Extreme Gradient Boosting for Decision Making

The final decision about subtitle display is made at the end of the pipeline by an Extreme Gradient Boosting Model (Chen and Guestrin, 2016). An optional bias, which is upstream of the sigmoid function, also allows additional regulation of the number of displayed subtitles during use.

4.4.1 Training Set

The dataset used for training and evaluation comprises a total of 39,367 words, corresponding to approximately five hours of video material. The data was manually labeled by both the author and additional annotators to determine the necessity of subtitle display for each word. The split into training and test sets was performed in an 80:20 ratio to ensure robust generalization and avoid overfitting. The following table provides an overview of the data distribution and the existing class imbalance:

category	**number**	**percent**
total dataset	39 367	100 %
training set	31 493	80,0 %
test set	7 874	20,0 %
True (Anzeigen)	3 416	8,7 %
False (Nicht-Anzeigen)	35 951	91,3 %
imbalance ratio (neg:pos)	10,52 : 1	

Table 2: Overview of training and test sets and class distribution.

To ensure the model's generalizability for different user profiles, the dataset was diversified regarding the audio and language comprehension levels (CEFR) selectable by users. The annotations were performed predominantly by the author, which particularly strengthens the representation of the C1 and C2 combination. The following table shows the distribution of words across the respective user preference combinations:

Table 3: Training set distribution by user preference (audio and language level).

audio level	language level	words	percent
B2	B1	6 629	16,8 %
B2	B2	3 507	8,9 %
C2	C1	22 134	56,3 %
C2	C2	7 097	18,0 %
total		39 367	100 %

4.4.2 Hyperparameters and Training Details

During the project, the use of Extreme Gradient Boosting (Chen and Guestrin, 2016) proved to be a more interpretable and efficient solution compared to neural networks. Hyperparameter optimization was performed using Bayesian optimization with a focus on maximizing the AUC-PR score. The final parameter configuration is:

- **Learning Rate:** 0.140
- **Maximum Depth:** 7
- **Subsample Ratio:** 0.923
- **Feature Subsample:** 0.719
- **L2 Regularization:** 7.951
- **Gamma:** 2.345
- **Minimum Leaf Weight:** 6.48
- **Positive Class Scaling:** 15.03

The model was trained with 1,000 trees and used early stopping after 50 rounds without improvement. Class imbalance was compensated through positive class scaling.

5 Results

This section presents the results of the developed pipeline and the XGBoost model and examines whether the hypothesis formulated at the beginning could be confirmed. The model was evaluated on the test set, which comprises 7,874 words (20% of the total dataset).

5.1 Model Performance Evaluation

The final evaluation of the XGBoost model on the test set provides the performance metrics shown in Table 4.

metric	F1-score	F2-score	balanced accuracy	ROC-AUC
Wert	0,4936	0,6122	0,8064	0,8927

Table 4: Performance metrics of the XGBoost model.

The F2-score of 0.6122, which weights recall more heavily than precision, indicates that the model has a good ability to identify actually needed subtitles (high recall). This is crucial for the project goal to ensure that users don't miss important information. The F1-score is noticeably

lower at 0.4936, indicating lower precision – the model also generates a relevant number of false-positive predictions (displays subtitles where they are not necessary). The balanced accuracy of 0.8064 shows solid average performance across both classes, even considering class imbalance. The ROC-AUC value of 0.8927 is relatively high and confirms that the model has good general ability to distinguish between the classes "Display" and "Non-Display".

5.2 Classification Report

The detailed classification report (Table 5) provides further insight into the performance for individual classes.

class	precision	recall	F1-score	support
false (non-display)	0,97	0,88	0,93	7191
true (display)	0,37	0,73	0,49	683
accuracy			0,87	7874
macro Avg	0,67	0,81	0,71	7874
weighted Avg	0,92	0,87	0,89	7874

Table 5: Detailed classification report of the XGBoost model.

The analysis of the classification report confirms the observations from the overall metrics. For class 0 (Non-Display), the model achieves excellent performance: the high precision (0.97) means that words classified as "Non-Display" are very likely to indeed not need to be displayed, while the recall (0.88) and F1-score (0.93) are also strong, indicating effective suppression of unnecessary subtitles. For class 1 (Display), the expected trade-off is evident: the recall is good at 0.73, so 73% of actually needed subtitles are recognized, which serves the project goal of not missing important information. However, precision is low at 0.37, meaning that many subtitles are falsely displayed (false positives) when the model suggests display. The low F1-score of 0.49 for this class reflects this discrepancy between good recall and weak precision.
The high overall accuracy (Accuracy: 0.87) and high weighted average (Weighted Avg F1: 0.89) are strongly influenced by performance on the dominant class 0. The macro average (Macro Avg F1: 0.71), however, which weights both classes equally, gives a better impression of average performance across both classes and is pulled down by the weaker performance on class 1. In summary, the model is very good at avoiding unnecessary subtitles but displays too many unneeded subtitles when identifying necessary ones. The focus on recall (via F2-score and class weighting) was achieved, but at the cost of precision for the "Display" class.

5.3 Feature Importance Analysis

An advantage of Gradient Boosting models is the ability to analyze the importance of individual features for model decisions. Table 6 shows the Feature Importance scores, which indicate how much each feature contributed across all trees.
The feature importance analysis shows that user-specified preferences (User Language Level) as well as intrinsic properties of words (Word Familiarity, Word Frequency) have the greatest influence on model predictions. User Audio Comprehension Level and Word Complexity follow at some distance. This aligns with the intuition that both subjective user assessment and objective difficulty levels and word frequency are crucial for determining whether a subtitle is needed.
The features Sentence Complexity, Speech Rate, Relative Voice Volume, Ambient Volume, Audio Complexity, and Word Importance show lesser but still significant influence on model decisions. The comparatively lower importance of Speech Rate, volume-based features (Ambient Volume, Relative Voice Volume), and Audio Complexity could be due to insufficient variance in these features in the used dataset to enable stronger differentiation, or that the calculation methods

feature	importance score
user language level	0.1743
word familiarity	0.1591
word frequency	0.1451
user audio comprehension level	0.0905
word complexity	0.0719
sentence complexity	0.0689
speech rate	0.0637
relative voice volume	0.0618
ambient volume	0.0565
audio complexity	0.0554
word importance	0.0527

Table 6: Feature importance of the XGBoost model.

do not optimally capture the relevant aspects. The lower importance of Sentence Complexity and Word Importance compared to pure word-based features suggests that for the current modeling task, complexity and frequency determined at the word level are more relevant than more complex sentence or context analyses. This feature importance analysis provides valuable insights into the model's functioning and can serve as a basis for future optimizations of feature extraction or targeted expansion of the dataset.

5.4 Influence of Training Set Size

To investigate the influence of available training set size on XGBoost model performance, the model was trained with different portions of the total dataset and evaluated on the remaining part. Figure 2 shows the development of F2-score as a function of the number of words in the training set.

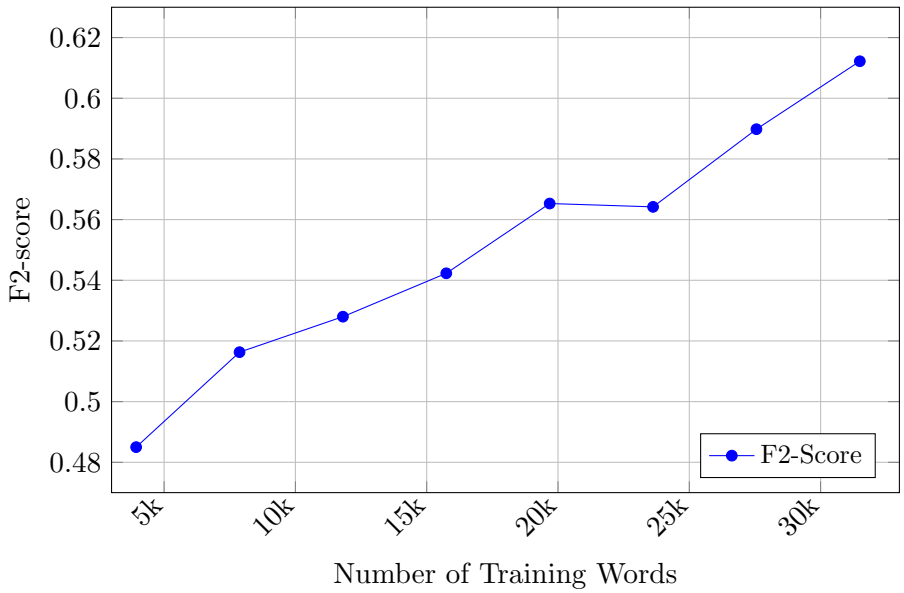


Figure 2: Development of F2-score (XGBoost) as a function of training set size.

The analysis of the relationship between training set size and model performance shows a clear positive trend. The F2-score improves from 0.485 when using only 10% of the data (approximately 3,900 words) to 0.612 at 80% of the data (approximately 31,500 words). The increase is observable over most of the range, with slight flattening and a minimal decline at 60% training

portion, before the score increases more strongly toward the end and reaches its highest value at the largest investigated training set size.
This development suggests that the XGBoost model benefits from a larger amount of data and that performance saturation with the currently available approximately 31,500 training words has not yet been reached. Further enlargement of the training set would very likely lead to additional performance improvements. For future work, evaluation with even larger amounts of data or the use of data augmentation techniques would be promising to further optimize model performance.

6 Results Discussion

The achieved results impressively demonstrate the fundamental feasibility and potential of adaptive subtitle systems. The central hypothesis that subtitles can be displayed on demand based on a combination of audio and text features as well as user preferences could be confirmed in essence. The developed XGBoost model is capable of significantly controlling the display frequency of subtitles.
However, the detailed performance analysis reveals a differentiated picture. The achieved F2-score of 0.6122 and the recall of 0.73 for the "Display" class (True) show that the model largely meets the set goal of reliably displaying important information. This minimizes the risk that users lack comprehension bridges. Conversely, however, there is the precision of only 0.37 for the same class. This means that a considerable portion of displayed subtitles were classified as unnecessary according to the annotation (false positives), which counteracts the goal of reducing visual distraction. The excellent precision of 0.97 for the "Non-Display" class (False) underscores the model's strength in effectively suppressing clearly unnecessary subtitles. The high ROC-AUC value (0.8927) also demonstrates the model's good general separation ability.
The feature importance analysis (Table 6) provides valuable insights: explicit user preferences (language and audio level) as well as basic word features (familiarity, frequency in video) dominate the decision. Features like speech rate or volume ratios have less influence, which could indicate possible limitations in feature extraction or insufficient variance of these features in the current dataset.
The investigation of training set size influence (Figure 2) shows a clear positive relationship: the F2-score increases continuously up to the maximum used data amount of 80% (approximately 31,500 words). This suggests that the model's performance potential has not yet been exhausted and that additional data can lead to significant improvements.
Despite the promising results, clear starting points for optimizations exist:

- **Training set expansion and diversification:** The current dataset is predominantly based on the author's annotations, leading to strong weighting of higher comprehensibility levels (C1/C2) (see Table 3). Expansion with data from users with different audio and language comprehension levels as well as from various genres (films, series, documentaries) is essential to increase the model's generalizability and robustness.

- **Precision improvement:** Reducing false positives for the "Display" class is a central goal. This could be achieved through refinement of existing features or development of new features.

- **Pipeline efficiency optimization:** The current processing time per video is considerable. Optimization of individual pipeline components, particularly through stronger GPU utilization and more efficient algorithms, is necessary for practical applicability.

7 Conclusion and Outlook

The present work has successfully developed and evaluated a system for adaptive subtitle control. The core question of whether subtitles can be dynamically faded in and out based on audio and text comprehensibility as well as user preferences could be answered positively. The developed pipeline, which uses an Extreme Gradient Boosting (XGBoost) model, represents a promising approach to overcome the traditional disadvantages of permanent or completely absent subtitles. The quantitative evaluation reveals the strengths and weaknesses of the current system. The focus on high recall (0.73 for class 1), supported by optimization on the F2-score (0.6122), ensures that users are shown important information with high probability. The downside is the low precision (0.37 for class 1), which leads to excessive subtitle display and limits the potential for reducing visual distraction. Nevertheless, the solid balanced accuracy (0.8064) and high ROC-AUC value (0.8927) demonstrate the model's fundamental ability to differentiate between necessary and unnecessary subtitles.
The original hypothesis regarding the sufficient representativeness of the chosen features must be considered differentially. While the features provide a basis for decision-making, the performance, particularly the precision limitation, suggests that the current feature representation still has optimization potential or that the model would benefit from more extensive data. The training set size analysis strongly underscores the latter point: performance increases up to the maximum available data amount, which strongly argues for acquiring additional annotated data. An interesting byproduct of the pipeline is the separation of voices and background noise using demucs. These separated audio tracks also enable another useful feature for improving comprehensibility: individual adjustment of relative volume between speaker voices and background ambience by users.
Clear directions emerge for future research and development work:

- **Data expansion and diversification:** This is the most promising path to performance improvement. A larger and more diverse dataset (more annotators, various user profiles, different genres and speaker styles) is essential.
- **Feature engineering and selection:** Investigation of additional or refined features (e.g., more complex linguistic features, consideration of longer contexts, more robust audio analyses) as well as critical evaluation of existing features could improve precision.
- **Investigation of open research questions:** Systematic analyses should address (a) what effects different genres (e.g., action film vs. drama) have on model performance, (b) how different distributions of language/audio levels in the training set affect performance, and (c) what performance is achieved on completely unknown language/audio levels.
- **User studies:** Subjective evaluations by end users are necessary to validate the actual improvement in viewing experience and comprehensibility and additionally to check whether using the system also results in a significant learning effect regarding English skills.

In summary, this work lays a solid foundation for adaptive subtitle systems. It demonstrates feasibility, identifies central challenges, and points out clear paths for future improvements. The developed approach is principally language-independent and could thus be transferred to other languages with appropriate adaptations.

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9 Support Services

The entire development of the pipeline as well as the neural network was performed completely independently. The conception, implementation, and evaluation were based exclusively on own ideas and approaches.
Various AI models were used to support the written elaboration and to verify concepts as well as program parts:

- **ChatGPT-4o/4o mini:** Writing the written work. Explanation of various procedures around ML/AI. Help with pipeline conception.
- **Gemini 2 Experimental:** Help with programming the code.
- **Perplexity (Claude 3.5 Sonnet/OpenAI o1-Preview):** See Gemini.